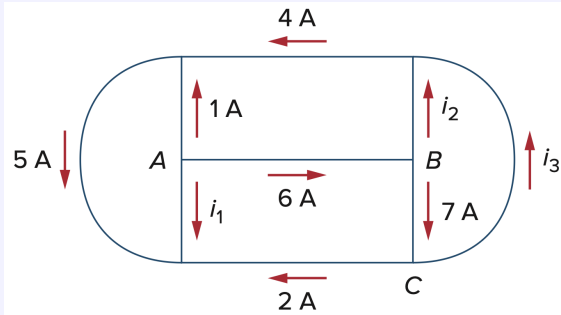


Exercise Sheet #2: DC 1: Circuit Calculations

Discussion: Oct 22, 2025

Exercise 1: KCL

Find i_1 , i_2 , and i_3 .

Solution

$$-1 \text{ A} - 6 \text{ A} - i_1 = 0$$

$$\Rightarrow i_1 = -1 \text{ A} - 6 \text{ A} = -7 \text{ A}$$

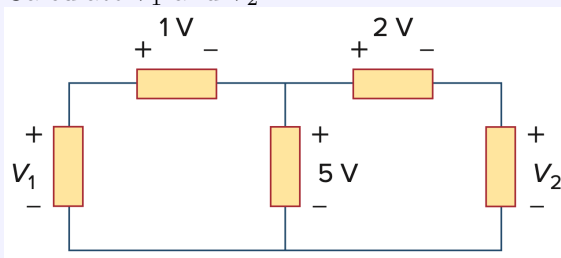
$$7 \text{ A} - 2 \text{ A} = 5 \text{ A} - i_3 = 0$$

$$\Rightarrow i_3 = 7 \text{ A} - 2 \text{ A} = 5 \text{ A}$$

$$i_2 + i_3 - 4 \text{ A} = 0$$

$$\Rightarrow i_2 = 4 \text{ A} - i_3 = -1 \text{ A}$$

Exercise 2: KVL

Calculate V_1 and V_2 .

Solution

$$1 \text{ V} + 5 \text{ V} - V_1 = 0$$

$$\Rightarrow V_1 = 1 \text{ V} + 5 \text{ V} = 6 \text{ V}$$

$$2 \text{ V} + V_2 - 5 \text{ V} = 0$$

$$\Rightarrow V_2 = 5 \text{ V} - 2 \text{ V} = 3 \text{ V}$$

Exercise 3: ...

...

Solution

...